IT3011 - Theory & Practices in Statistical Modelling

One Population Mean

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Chi – square Distribution

Chi-square distributions are used in many inferential problems, for example, in inferential problems dealing with variance. It is a probability distribution which consists of a single parameter. If n is a positive integer, then the parameter is called the degrees of freedom. The mean & variance of a Chi-square distribution are n and $2n$ respectively.

Let $x_1, x_2, x_3, \ldots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$. The we know that,

$$Z_i = \frac{(X_i - \mu)}{\sigma} \sim N(0,1\ )$$

Then

$$Z_i^2 \sim \chi^2_{(1)}$$

Then

$$\sum_{i=1}^{n} Z_i^2 \sim \chi^2_{(n)}$$

Chi – square Distribution

Chi-square distributions are used in many inferential problems, for example, in inferential problems dealing Let $x_1, x_2, x_3, \ldots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$. The we know that,

1. When μ is known,

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2_{(n)}$$

2. When μ is unknown,

$$\sum_{i=1}^{n} \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi^2_{(n-1)}$$

Since $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{X})^2$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$

t Distribution

If Y and Z are independent random variables, Y has a Chi-square distribution with n degrees of freedom and $\mathrm{Z} \sim N(0,1)$, then

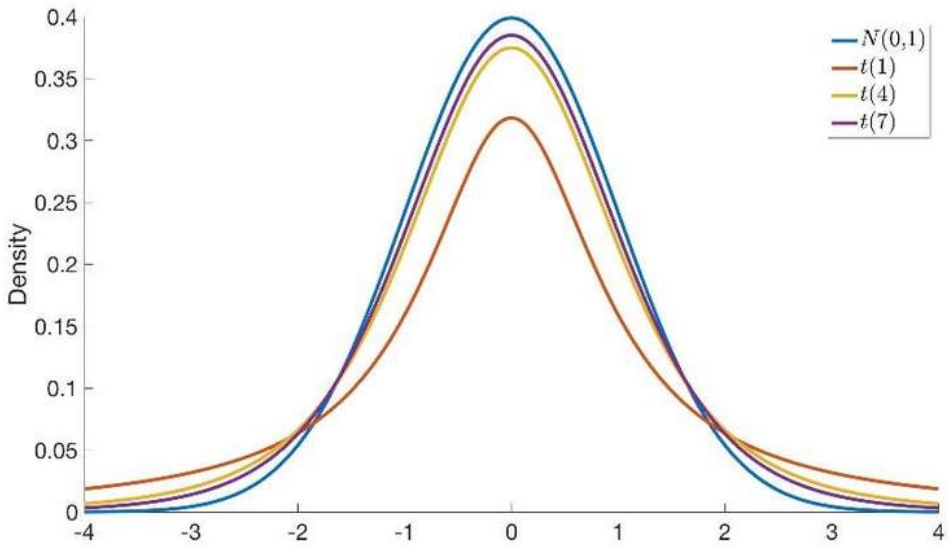
$$T = \frac{Z}{\sqrt{\frac{Y}{n}}}$$

is said to have a t distribution with n degrees of freedom. That is denoted by $T \sim t_{(n)}$.

t Distribution

The t distribution is similar to the standard normal distribution, but the height of the t distribution differs depending on the sample size. **t distribution is used when we don't know the population standard deviation and when we have small sample sizes (Say n<30).** When the sample size increases, the t distribution approaches the standard normal distribution.

The height of the t distribution is determined by the number of degrees of freedom.



t Distribution

Sample mean and sample variance are independent.

$$\mathrm{Z} = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \sim N(0,1\ )$$

We know that,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

So

$$\left(\frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}}\right) \Bigg/ \sqrt{\frac{\frac{(n-1)S^2}{\sigma^2}}{n-1}} = \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \sim t_{(n-1)}$$

F Distribution

Let U and V be Chi-squared random variables with n_1 and n_2 degrees of freedom respectively. If U and V are independent.

$$F = \frac{U/n_1}{V/n_2} \sim F(n_1, n_2)$$

Now consider two different populations with different population variances.

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} \sim \chi^2_{(n_1-1)} \; and \; \frac{(n_2 - 1)S_2^2}{\sigma_2^2} \sim \chi^2_{(n_2-1)}$$

So,

$$\frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

The Pivotal Method of Finding the Confidence Interval

Pivotal Quantity

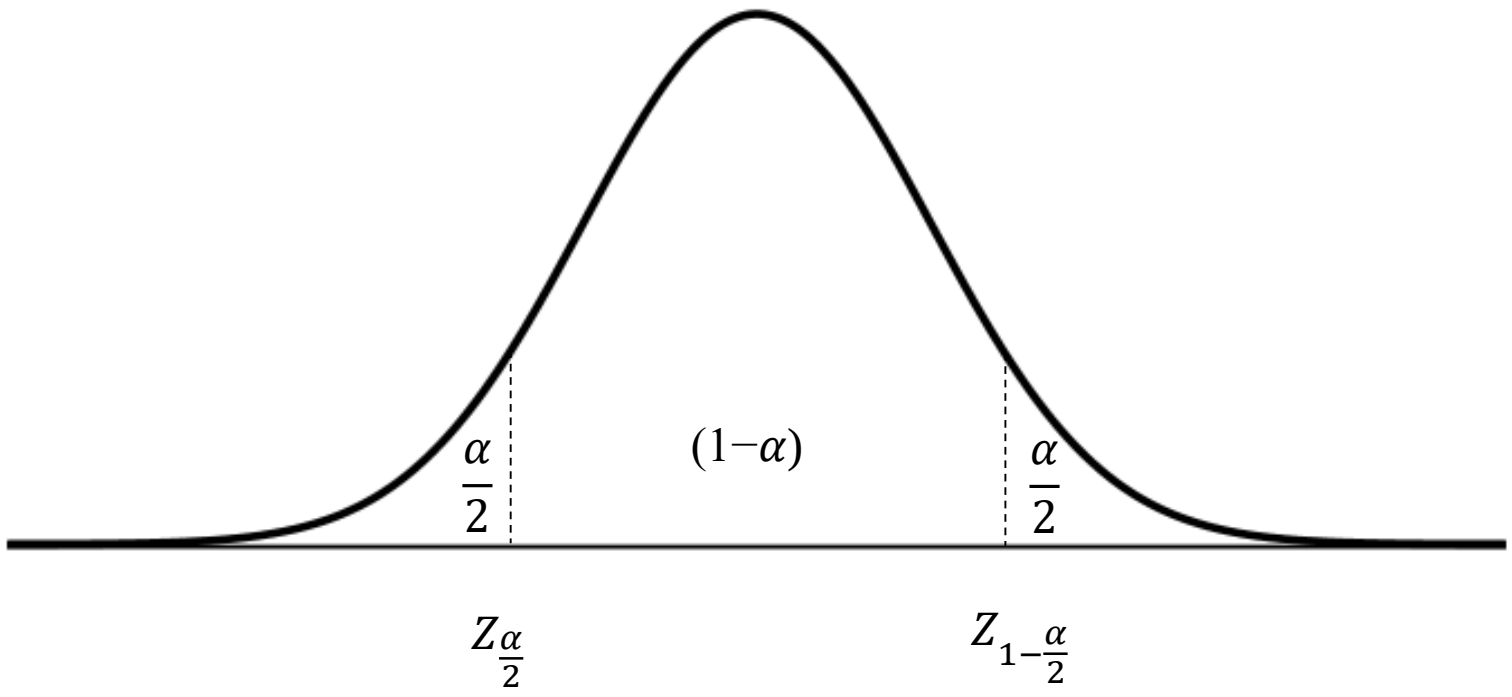
It is a function of the sample statistics and the unknown parameter where the unknown parameter is the only unknown quantity

It’s probability distribution does not depend on the unknown parameter

$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is known

In a random sample $x_1, x_2, x_3, \ldots, x_n$ from $N(\mu, \sigma^2)$, $\bar{X} \sim N\left(\mu, {}^{\sigma^2}/_{n}\right)$.

By standardizing $\bar{X}$, $\mathrm{Z} = \frac{(\bar{X}-\mu)}{{}^{\sigma}/_{\sqrt{n}}} \sim N(0,1)$



$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is known

We have to create,

$$P(L \leq \mu \leq U) = 1 - \alpha$$

For creating that we use the pivotal quantity: $\mathrm{Z} = \frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}}$

So,

$$P\left(Z_{\frac{\alpha}{2}} \leq \frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \leq Z_{(1-\frac{\alpha}{2})}\right) = P\left(-Z_{(1-\frac{\alpha}{2})} \leq \frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \leq Z_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha$$

By simplifying,

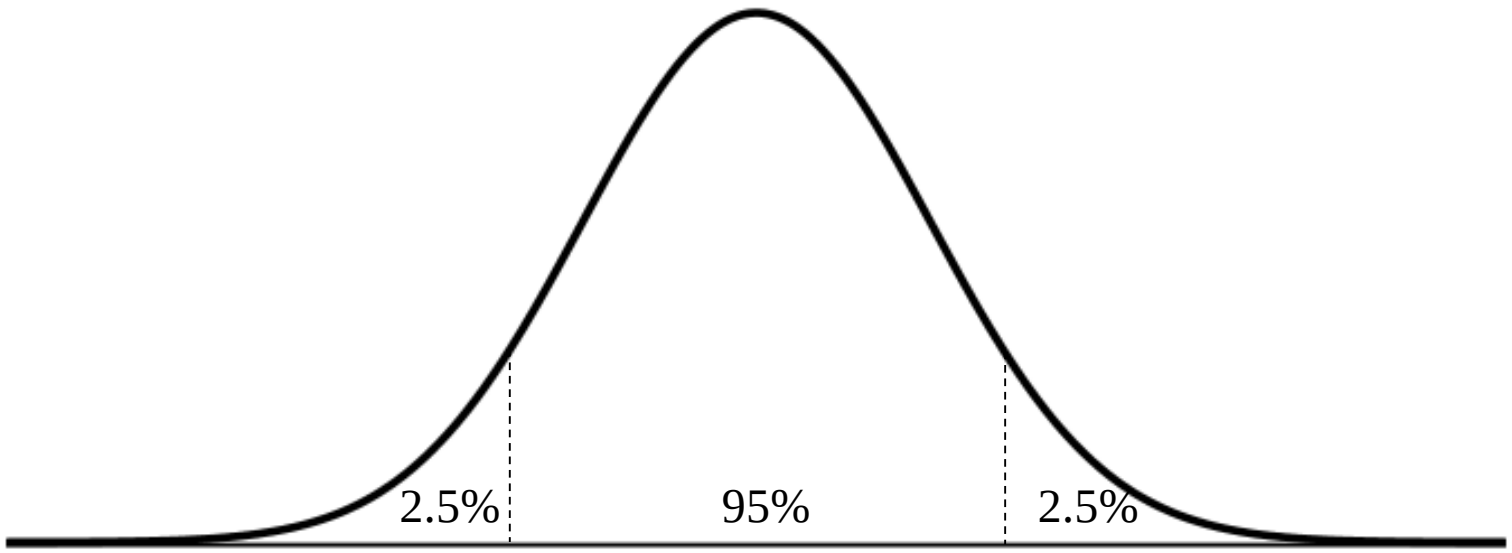
$$P\left(\bar{X} - \sigma/\sqrt{n}\, Z_{(1-\frac{\alpha}{2})} \leq \mu \leq \bar{X} + \sigma/\sqrt{n}\, Z_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha$$

Therefore a $(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - \sigma/\sqrt{n}\, Z_{\left(1-\frac{\alpha}{2}\right)} \quad , \quad \bar{X} + \sigma/\sqrt{n}\, Z_{(1-\frac{\alpha}{2})}\right]$$

Ex-

A random sample of 12 items is selected from a normal population with variance 4. Sample mean is 10.4. Find the 95% confidence interval for the population mean.

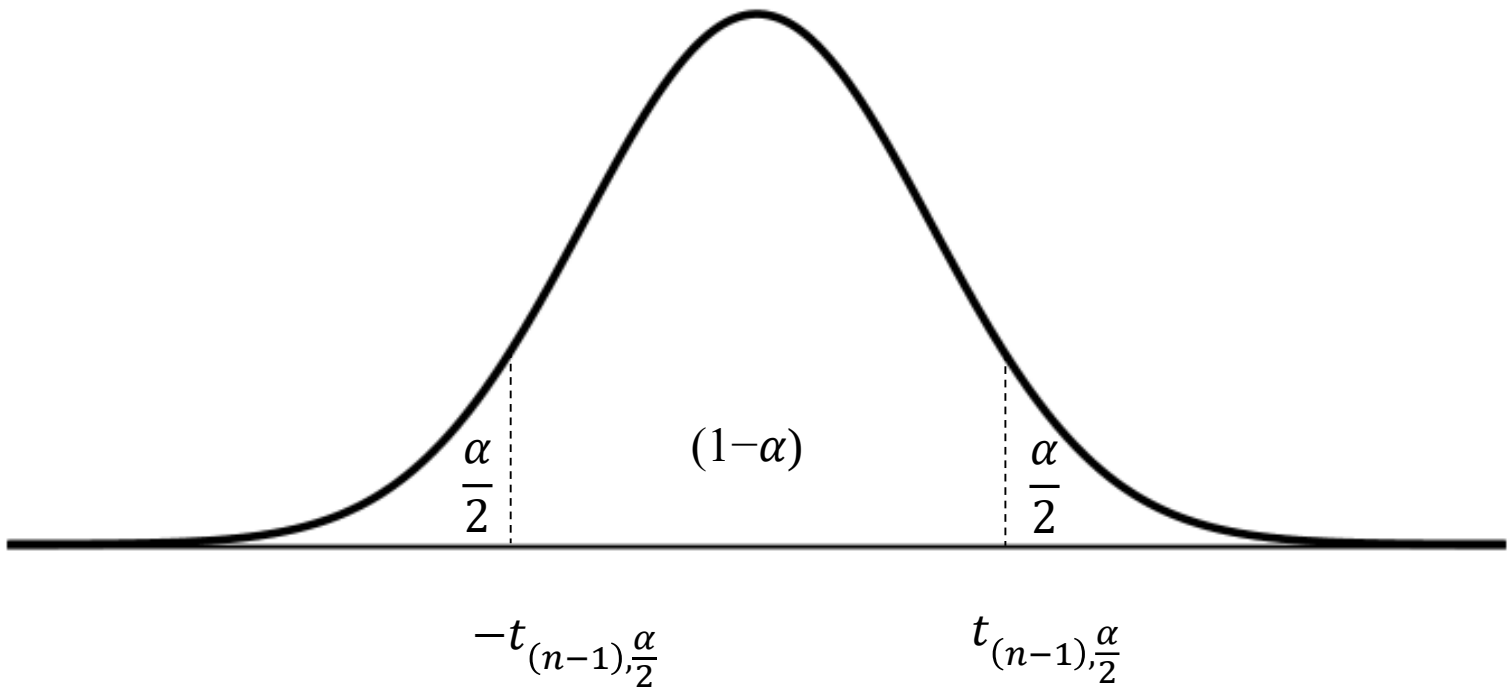


$Z_{2.5\%} = -1.96$ $\qquad$ $Z_{97.5\%} = 1.96$

$$\left[10.4 - {}^{2}/_{\sqrt{12}}\, Z_{2.5\%} \quad , 10.4 + {}^{2}/_{\sqrt{12}}\, Z_{2.5\%}\right] = [9.26\, , 11.52]$$

$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is unknown

Suppose we have a random sample $x_1, x_2, x_3, \ldots, x_n$ from $N(\mu, \sigma^2)$ where σ^2 is unknown. Lets focus about t distribution. Consider $\mathrm{T} = \frac{(\bar{X}-\mu)}{S/\sqrt{n}} \sim t_{(n-1)}$



$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is unknown

We have to create,

$$P(L \leq \mu \leq U) = 1 - \alpha$$

For creating that we use the pivotal quantity: $\mathrm{T} = \frac{(\bar{X}-\mu)}{S/\sqrt{n}} \sim t_{(n-1)}$

So,

$$P\left(-t_{(n-1),\frac{\alpha}{2}} \leq \frac{(\bar{X}-\mu)}{S/\sqrt{n}} \leq t_{(n-1),\frac{\alpha}{2}}\right) = 1 - \alpha$$

By simplifying,

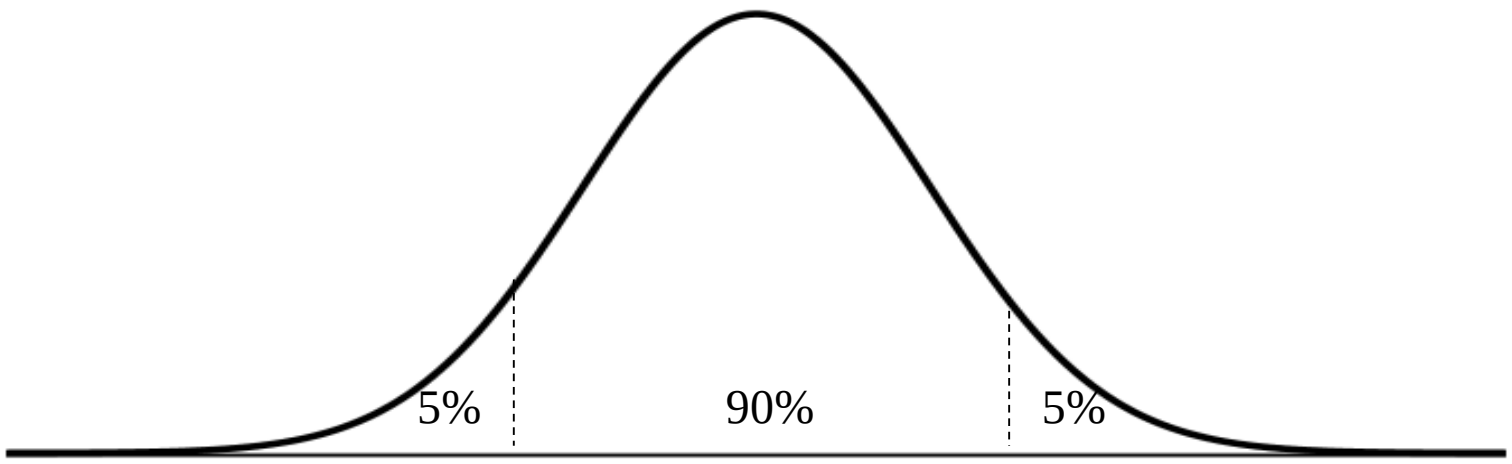
$$P\left(\bar{X} - S/\sqrt{n}\, t_{(n-1),\frac{\alpha}{2}} \leq \mu \leq \bar{X} + S/\sqrt{n}\, t_{(n-1),\frac{\alpha}{2}}\right) = 1 - \alpha$$

Therefore a $(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - S/\sqrt{n}\, t_{(n-1),\frac{\alpha}{2}} \quad , \quad \bar{X} + S/\sqrt{n}\, t_{(n-1),\frac{\alpha}{2}}\right]$$

Ex-

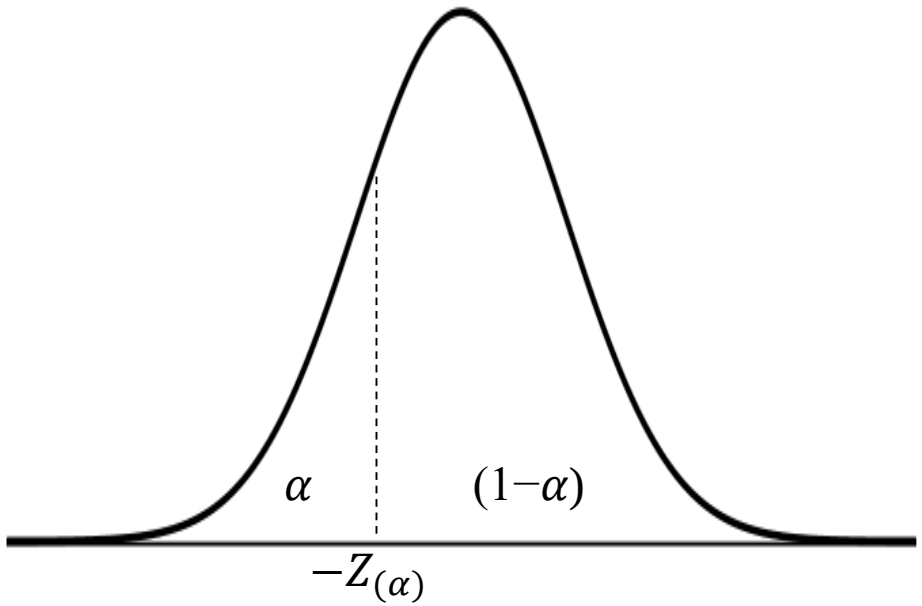
Let X equal the amount of butterfat in pounds produced by a typical cow during a 105-day milk production period between her first and second calves. Assume that the distribution of X is normal but the population variance is unknown. Construct a 90% confidence interval for population mean when the sample mean is 507.7 and sample standard deviation is 89.75. (n=20)



$$-t_{(20-1),5\%} = -1.729 \qquad t_{(20-1),5\%} = 1.729$$

$$\left[507.5 - {}^{89.75}/_{\sqrt{20}}\, t_{(20-1),5\%}, 10.4 + {}^{89.75}/_{\sqrt{20}}\, t_{(20-1),5\%}\right] = [472.801\,, 542.199]$$

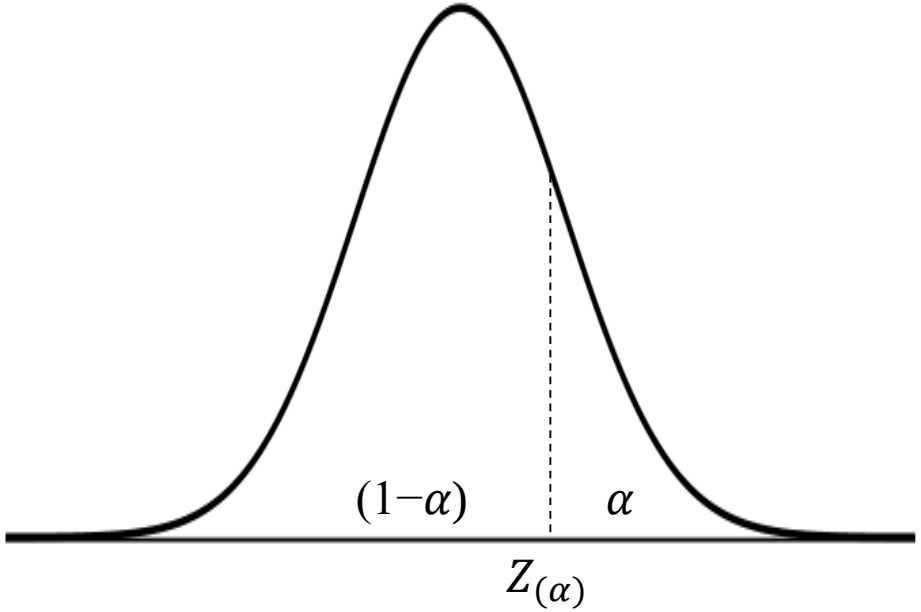
One Sided Confidence Interval for mean (μ) when variance is known



$$P\left(\frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \geq -Z_{(\alpha)}\right) = 1-\alpha$$

$$P\left(\mu \leq \bar{X} + \sigma/\sqrt{n}\, Z_{(\alpha)}\right) = 1-\alpha$$

$$\left[-\infty\,, \quad \bar{X} + \sigma/\sqrt{n}\, Z_{(\alpha)}\right]$$



$$P\left(\frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \leq Z_{(\alpha)}\right) = 1-\alpha$$

$$P\left(\mu \geq \bar{X} - \sigma/\sqrt{n}\, Z_{(\alpha)}\right) = 1-\alpha$$

$$\left[\bar{X} - \sigma/\sqrt{n}\, Z_{(\alpha)} \quad ,\infty\right]$$

Type I Versus Type II Errors

When testing hypotheses there are essential two errors that can be made.

1. **Type I error** - Rejecting the null hypothesis when the null is correct actually.
2. **Type II error** - Not rejecting the null hypothesis when the null is not correct actually.

Decision	Reality	
	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct
Do not reject H_0	Correct	Type II error

$$P(Type\ I\ error) = \alpha$$

$$P(Type\ II\ error) = \beta$$

Significance Level

When we conduct a hypothesis test we will always have to specify a significance level. We basically have to answer the question – “What is the probability of a Type I error we are willing to live with?”.

We cannot generally quantify the probability of a type II error. To avoid making Type ii errors we will always say “Do not reject the null hypothesis” instead of “Accept the null hypothesis”.

Popular Significance Levels

0.1 0.05 0.01

Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(1) Classical Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Find the appropriate test statistic and its distribution.
- Compute the test statistic under H_0.
- Define the rejection region according to the given significance level.
- If the calculated test statistic falls in the rejection region, reject H_0 and if not, do not reject H_0.

Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(2) Confidence Interval Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Construct appropriate confidence interval for specified level of significance under H_0.
- If the value of the parameter under H_0 lies within the confidence interval, we do not reject H_0. If the value of the parameter under H_0 lies outside the confidence interval we reject H_0.

Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(3) P-Value Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Find the appropriate test statistic and its distribution.
- Compute the P-value based on computed value of the test statistic under H_0.
- If the P-value is less than the significance level, we reject H_0. Otherwise we do not reject H_0.

Tests About Mean : Z – test when population variance is known

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n. **This is a two tailed test**.

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

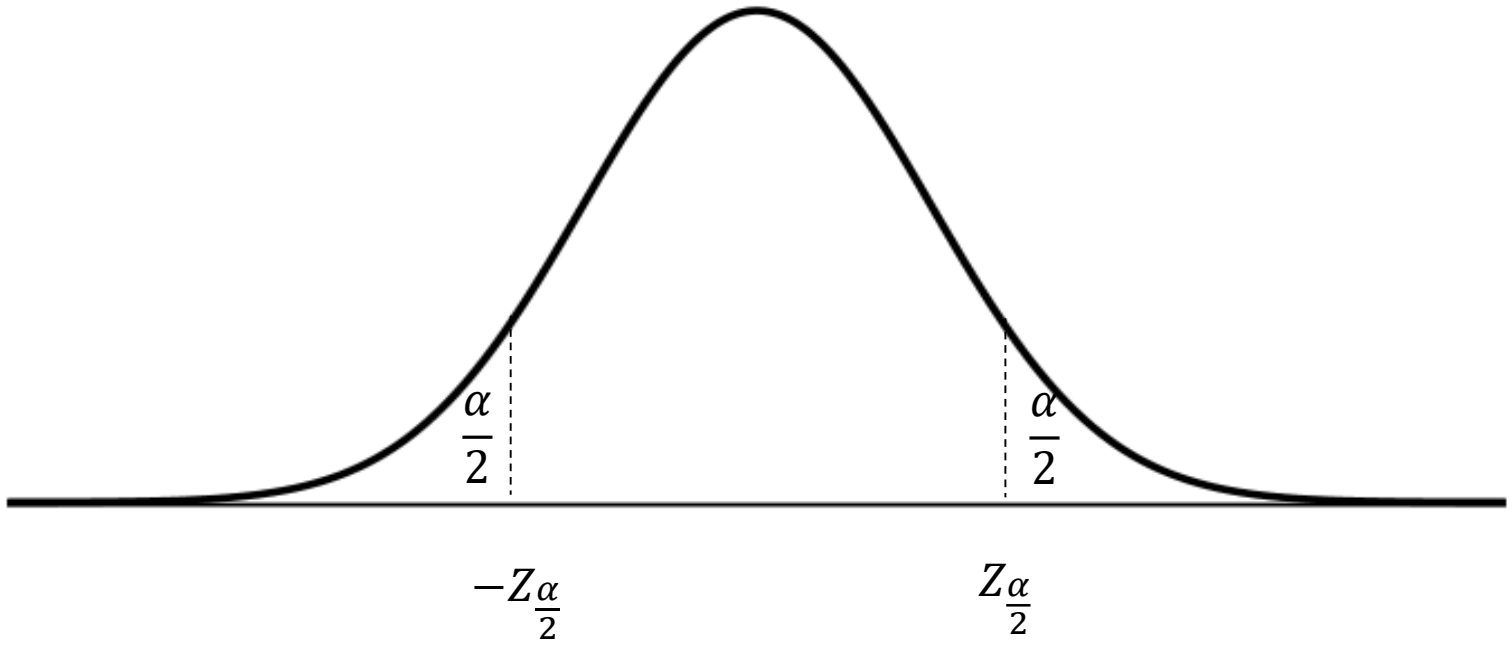
$$\mathrm{Z} = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \sim N(0,1)$$

Now compute the values of the test statistic under H_0. Then,

$$\mathrm{Z}_{cal} = \frac{(\bar{X} - A)}{\sigma/\sqrt{n}}$$

Tests About Mean : Z – test when population variance is known

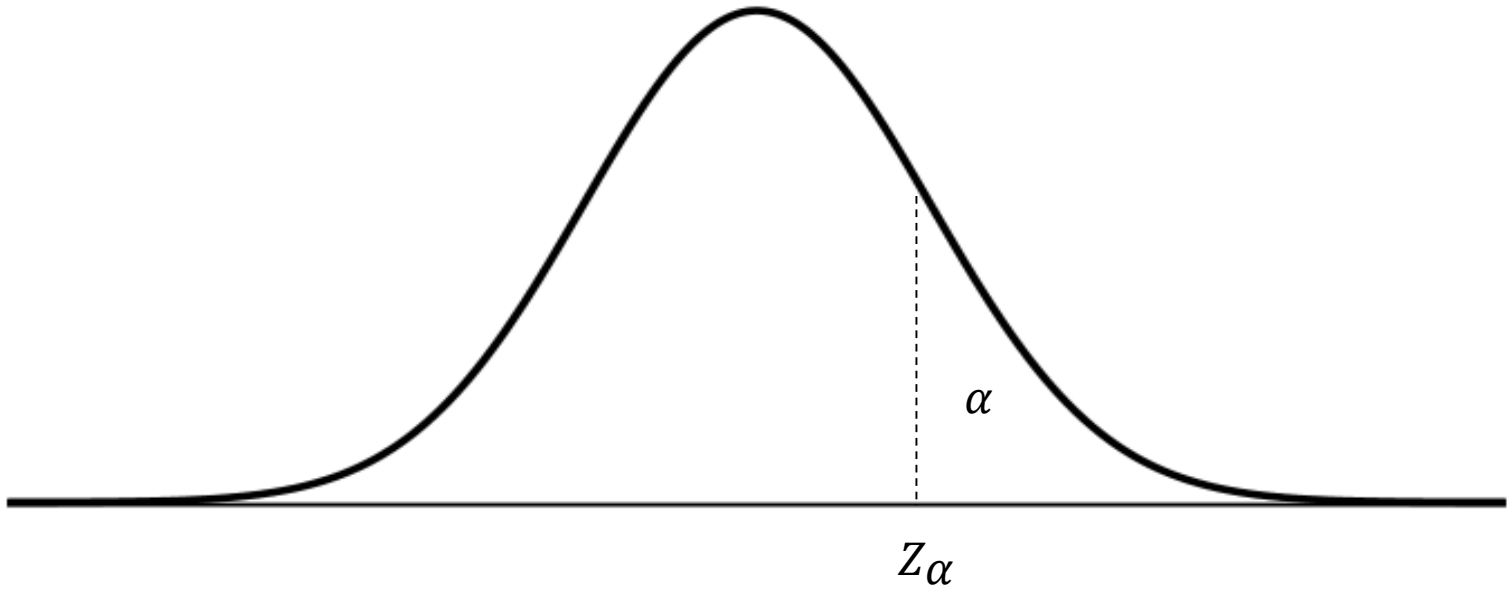
Since this is a two tail test,



If $Z_{cal} > Z_{\frac{\alpha}{2}}$ or $Z_{cal} < -Z_{\frac{\alpha}{2}}$ we reject H_0

Tests About Mean : Z – test when population variance is known

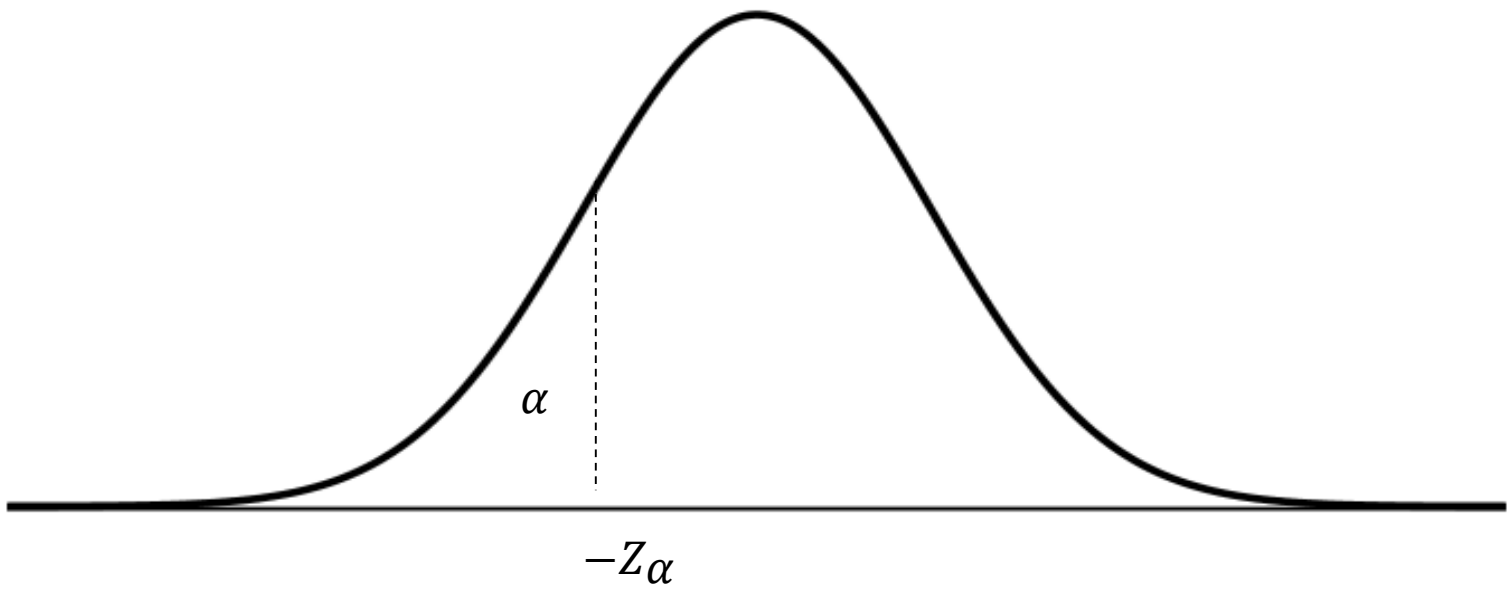
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $Z_{cal} > Z_\alpha$ we reject H_0

Tests About Mean : Z – test when population variance is known

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $Z_{cal} < -Z_\alpha$ we reject H_0

Ex-

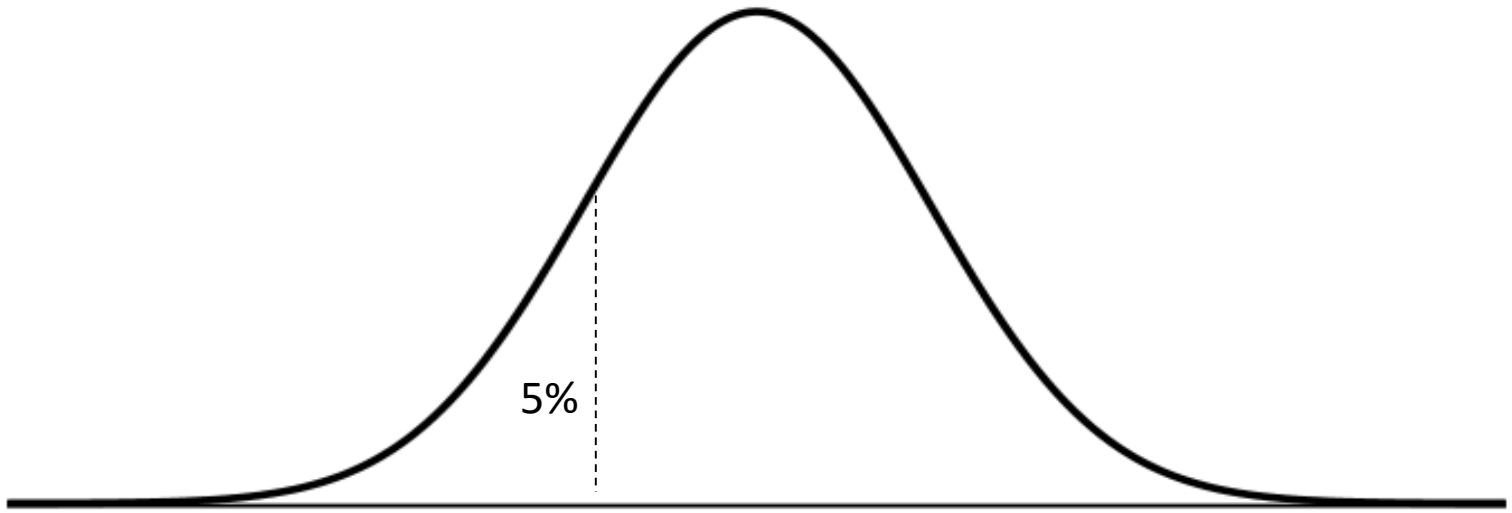
Boys of a certain age are known to have a mean weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. 25 boys from the home are weighed and found to have a mean weight of 80.94 pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data what should be concluded concerning the complaint? (Significance level 5%)

$$H_0 : \mu = 85$$

$$H_1 : \mu < 85$$

Under H_0 ,

$$\mathrm{Z}_{cal} = \frac{(80.94 - 85)}{11.6/\sqrt{25}} = -1.75$$



$-Z_{5\%} = -1.645$

$Z_{cal} = -1.75 < -Z_{5\%}$,so we reject H_0 at 5% level of significance.

Conclusion **– There is an evidence to say that, the boys living in a municipal children's home are underfed at 5% level of significance.**

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Tests About Mean : t – test when population variance is unknown

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n. **This is a two tailed test**.

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

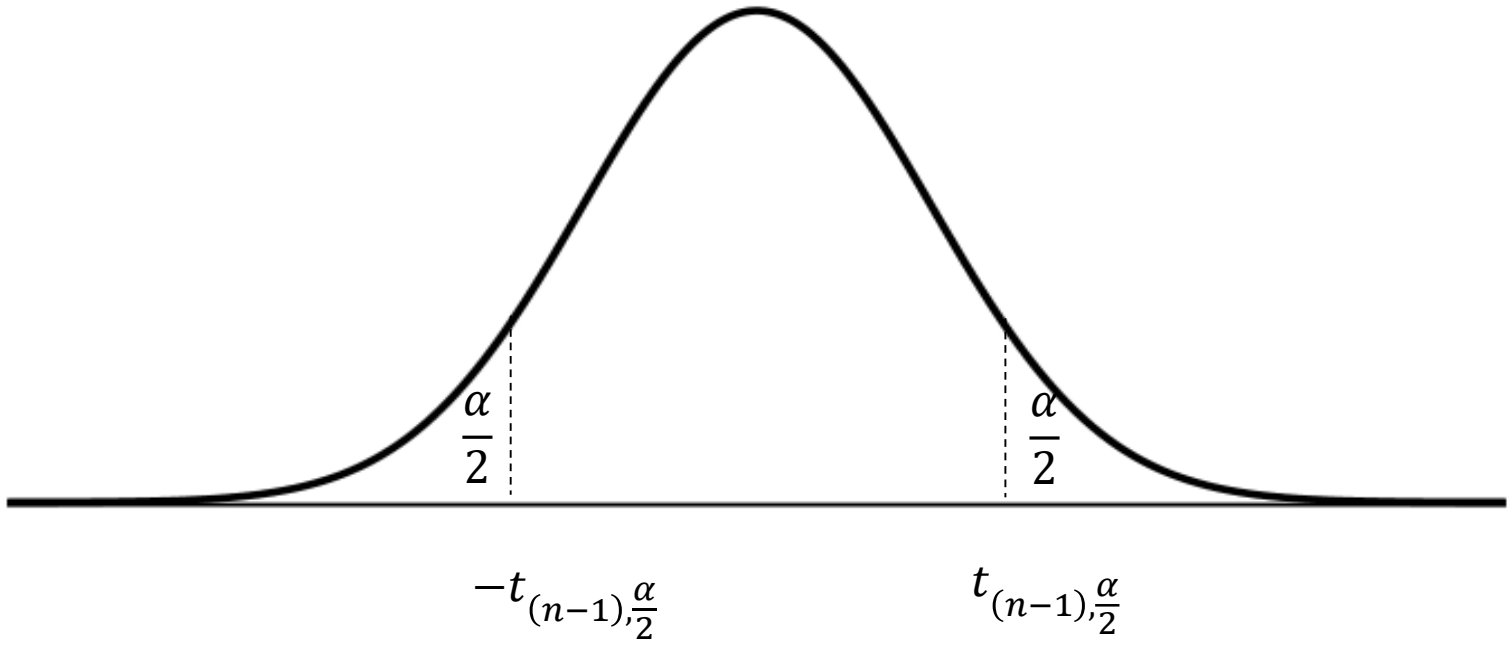
$$\mathrm{T} = \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \sim t_{(n-1)}$$

Now compute the values of the test statistic under H_0. Then,

$$\mathrm{t}_{cal} = \frac{(\bar{X} - A)}{S/\sqrt{n}}$$

Tests About Mean : t – test when population variance is unknown

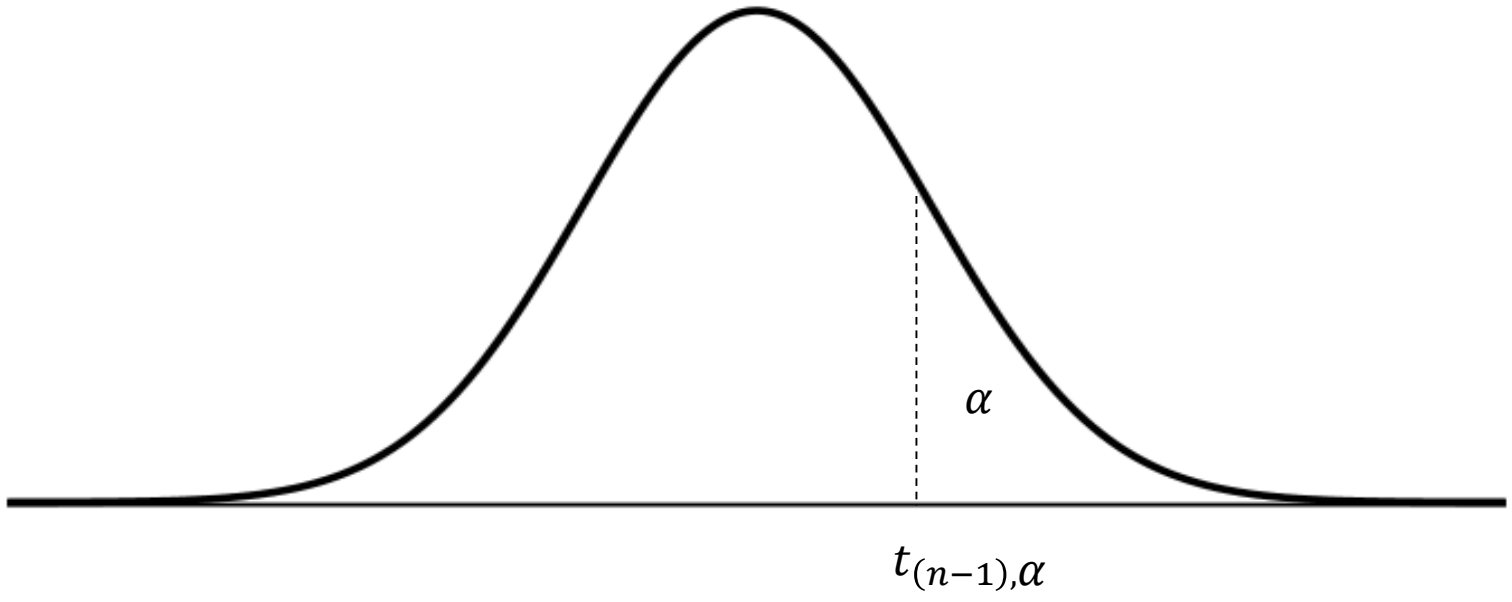
Since this is a two tail test,



If $t_{cal} > t_{(n-1),\frac{\alpha}{2}}$ or $t_{cal} < -t_{(n-1),\frac{\alpha}{2}}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

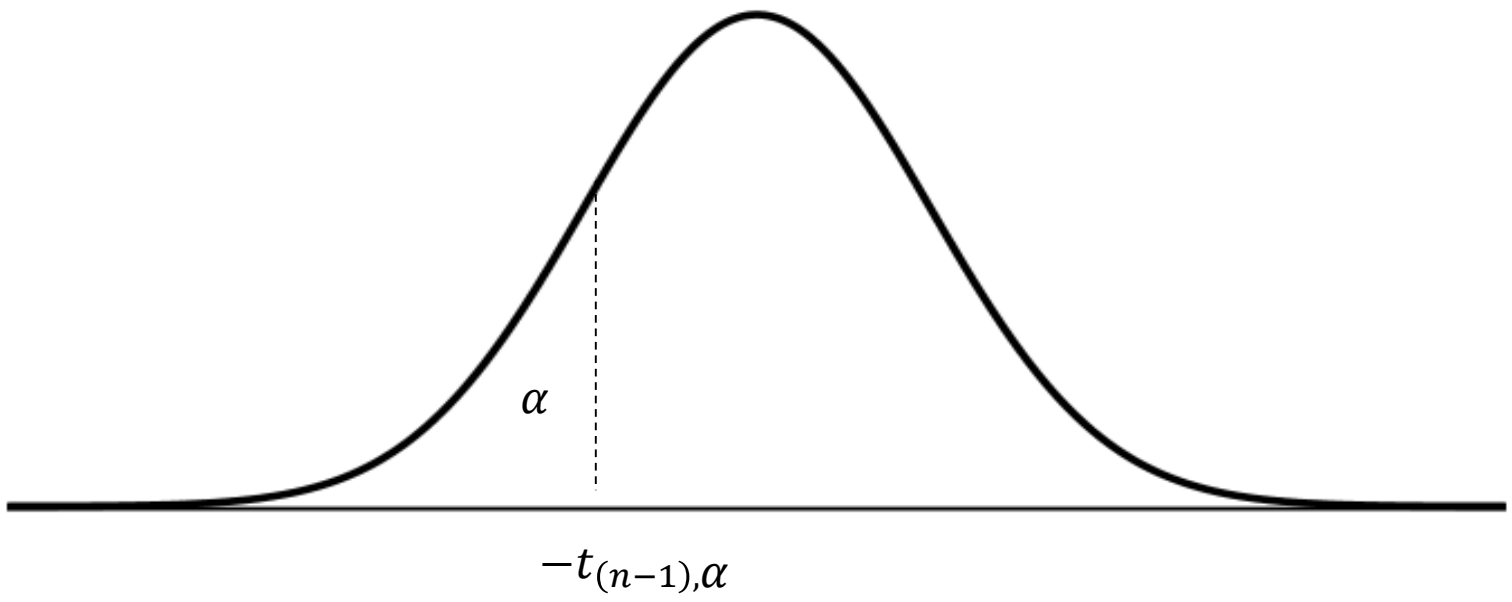
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $t_{cal} > t_{(n-1),\alpha}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $t_{cal} < -t_{(n-1),\alpha}$ we reject H_0

Ex-

Five readings of the resistance in Ohms (Ω) of a piece of wire gave the following results.

$$1.15, 1.49, 1.54, 1.52, 1.54$$

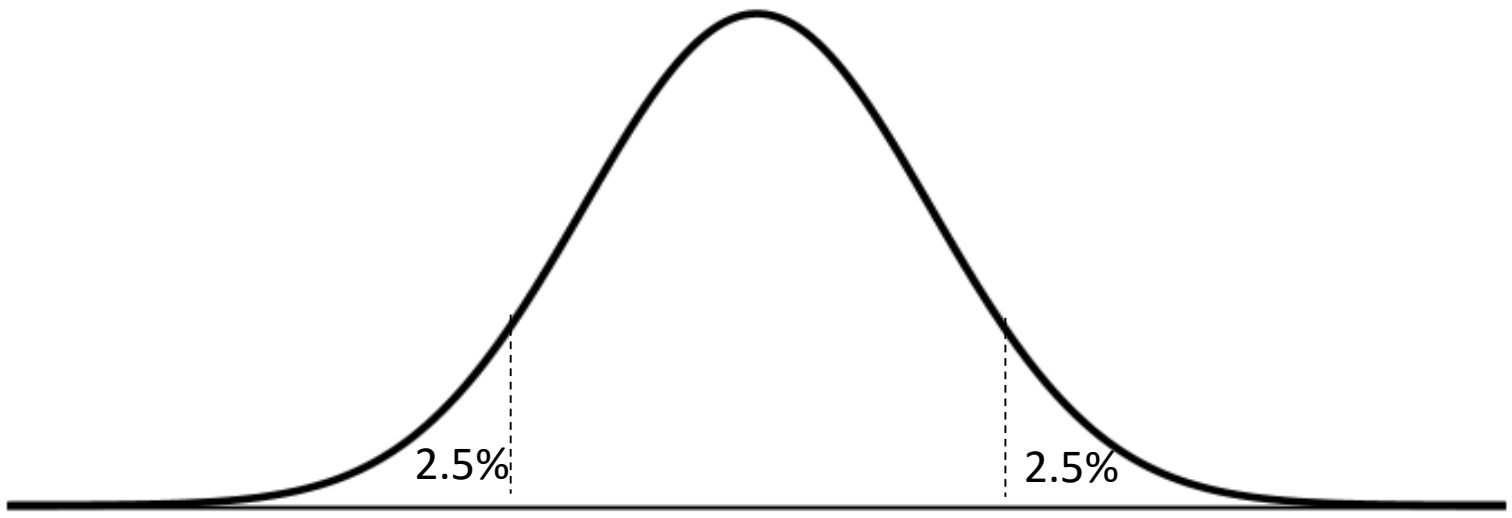
If the wire is pure silver, its resistance should be 1.5Ω. Test at 5% level of significance whether the wire is pure silver. Assume the normality of the readings.

$$H_0 : \mu = 1.5$$

$$H_1 : \mu \neq 1.5$$

Under H_0 ,

$$\mathrm{t}_{cal} = \frac{(1.448 - 1.5)}{0.15/\sqrt{5}} = -0.78$$



$-t_{(4),2.5\%} = -2.7763 \qquad t_{(4),2.5\%} = 2.7763$

$\mathrm{t}_{cal} = -0.78 > -t_{(4),2.5\%}$,so we do not reject H_0 at 5% level of significance.

Conclusion - **There is an evidence to say that the wire is pure silver at 5% level of significance.**

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

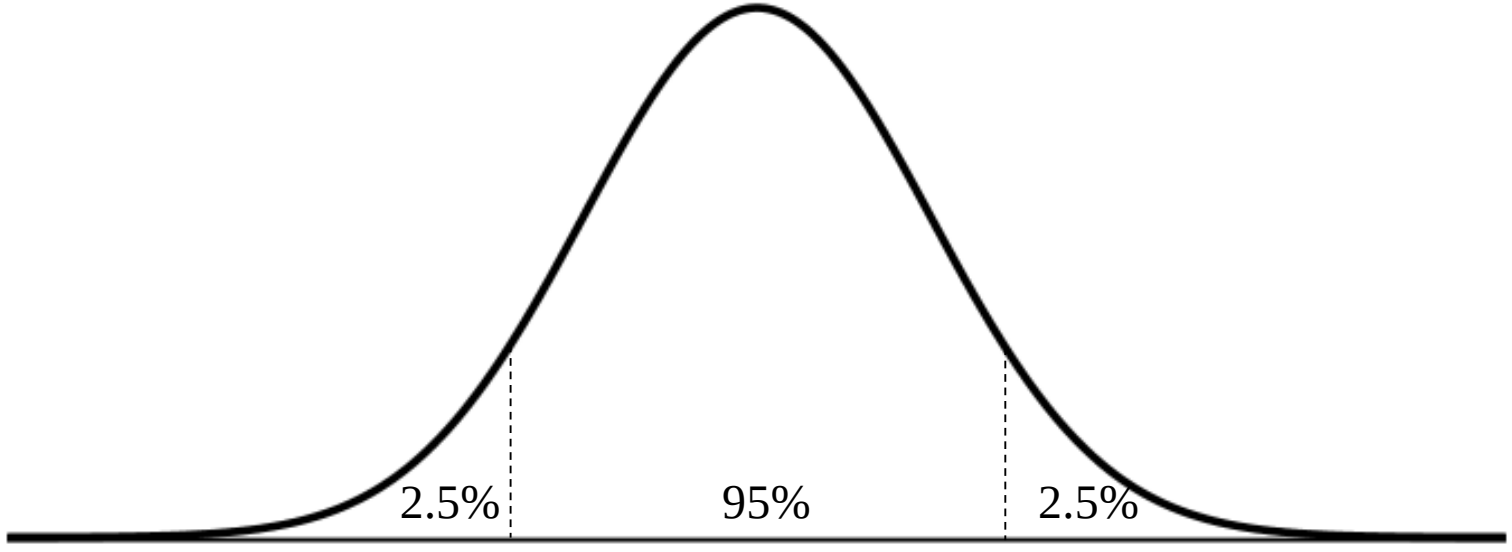
Confidence Interval Approach : Ex-

A random sample of 12 items is selected from a normal population with variance 4. Sample mean is 10.4. Test the following hypothesis at 5% level of significance.

$$H_0 : \mu = 15$$

$$H_1 : \mu \neq 15$$

Let's create the confidence interval for 5% level of significance under H_0. 5% significance means 95% confidence should be there.



$Z_{2.5\%} = -1.96$ $\qquad$ $Z_{7.5\%} = 1.96$

$$\left[10.4 - {}^{2}/_{\sqrt{12}} Z_{2.5\%} \quad , 10.4 + {}^{2}/_{\sqrt{12}} Z_{2.5\%}\right] = [9.26\,, 11.52]$$

The parameter under H_0 lies outside the confidence interval. So H_0 will be rejected at 5% level of significance.

Conclusion - **There is an evidence to say that the population mean is not equal to 15 at 5% level of significance.**

P-Value Approach : Ex-

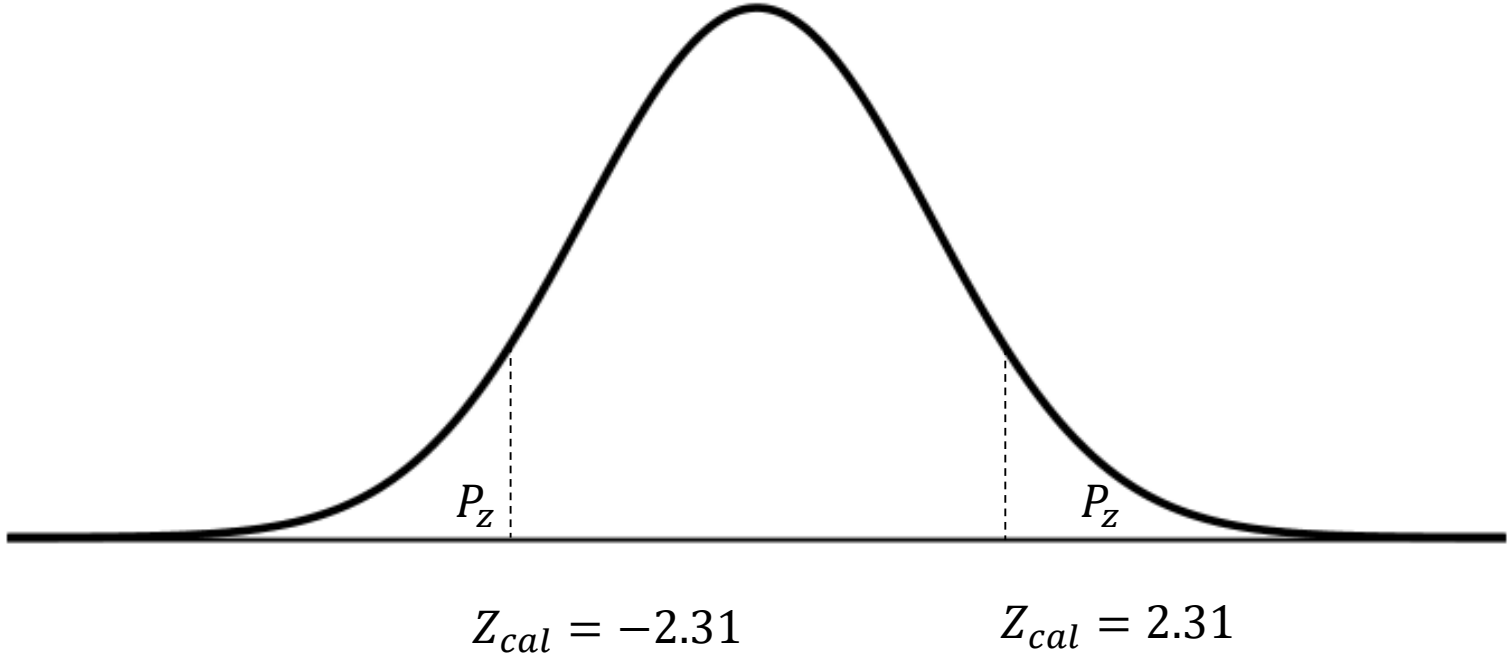
The average weight of all residents in town ABC is 168 lbs and the standard deviation is 3.9 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5. Test whether her idea is true or false at 5% level of significance.

$$H_0 : \mu = 168$$

$$H_1 : \mu \neq 168$$

Compute the test statistic under H_0.

$$\mathrm{Z}_{cal} = \frac{(169.5 - 168)}{3.9/\sqrt{36}} = 2.31$$



$$P_z = 0.01044$$
$$\boldsymbol{P - Value} = 2 \times P_z = 0.02088$$
$$\boldsymbol{P - Value} < Significance\ Level = 0.05$$

So H_0 will be rejected at 5% level of significance.

Conclusion - **There is an evidence to say that the nutritionist's idea is true at 5% level of significance.**